

Descriptive Written Test

25

① Smart Home Automation System

19/20

An Intelligent Agent is a system that perceives its important environment through sensors and take actions through actuators to achieve goals.

In a Smart home, sensors such as motion detectors, temperature, cameras, sensors and light sensors to collect information while devices like lights, air conditioners, heaters, alarms and smart locks perform actions.

Types of Agents(1) Simple Reflex Agent

A simple Reflex agent makes decisions only based on the current percept using predefined condition-action rules. It has no memory of previous events.

Working in the Smart Home:

If motion is detected → Turn on the lights.

If temperature is above 30°C → switch on the Air Conditioner.

If the door opens at night → Activate the security alarm.

Advantages

Fast response, Easy to implement, low computational cost

Disadvantages

Cannot Remember, previous events, cannot learn user habits, perform poorly in changing environment.

(ii) Model-Based Reflex Agent

A Model-Based Agent maintains an internal model of the environment and remembers previous states before making decisions.

Working in the Smart home:

- It remembers whether someone is already inside the room.
- Keeps track of indoor temperature changes.
- Detects if a window was left open while the AC is running.

Advantages

- Handles partially observable environments, makes better decisions using memory, improves energy efficiency.

Disadvantages

- More complex than simple Reflex Agents, requires additional memory.

(iii) Goal-Based Agent

A Goal-Based Agent selects actions that help achieve predefined goals.

Goals in Smart home

- Maintain a comfortable room temperature.
- Minimize electricity consumption.
- Keep the house secure.
- Reduce unnecessary lighting.

The Agent evaluates different possible actions and chooses the one that best achieves the goal.

Advantages:

Makes intelligent decisions, can evaluate multiple action paths.

Disadvantage

Requires planning.

Decision-making may take more time.

(IV) Utility-Based Agent

A utility-based Agent chooses the action that provides the highest overall benefit by considering multiple factors simultaneously.

Working in a smart home:

The Agent balances:

User Comfort, energy consumption, security and cost.

Advantages

Optimizes multiple objectives, Makes balanced decisions, produces the best overall performance.

Disadvantage

More computationally expensive.

Best choice

A Hybrid Agent Combining Model-Based, Goal based and utility based approaches is the most effective.

Justification

Model-Based remembers occupancy and environmental conditions.

Goal-Based ensures comfort, security and energy-saving objectives are achieved.

Although, each Agent has specific strengths, a Hybrid Agent combining all three is most-suited for a smart home because it can remember past events, multiple goals and optimize comfort.

②

Robot Placed in An Unknown Maze

The robot starts without any information about the maze. It must explore the environment, avoid dead ends, and eventually reach the exit.

(1) Uniform Cost Search (UCS)

Uniform Cost Search expands the node with the lowest cumulative path cost first.

Working in the Maze

- (1) Robot starts at the entrance.
- (2) All possible paths are inserted into a priority queue.
- (3) The path with the minimum cost is expanded.
- (4) process continues until the exit is found.

If one path cost 8 units, and another costs 5 units, UCS always explores the path costing 5 units.

Advantage

Finds the least cost path.

Complete.

Optimal when costs are non-negative.

Disadvantage

Requires large memory.
slow for large mazes.

Complexity

$$\text{Time: } O(b^l + [C^*/\epsilon])$$

$$\text{Space: } O(b^l + [C^*/\epsilon])$$

where,

$b \rightarrow$ branching factor

$C^* \rightarrow$ optimal path cost

$\epsilon \rightarrow$ minimum step cost.

(11) Iterative Deepening Search (IDS)

IDS repeatedly performs Depth-limited Search with increasing depth until the goal is found.

Working in the Maze

- Search depth 0
- Then depth 1
- Then depth 2
- Continue increasing until the exit is reached.

Advantages

Uses very little memory.

Complete.

Suitable when optimal path solution depth is unknown.

Disadvantages

Repeats exploration of upper levels.

Not optimal when path costs differ.

Complexity:

Time: $O(b^d)$

Space: $O(bd)$

where,

$b \rightarrow$ branching factor

$d \rightarrow$ depth of the solution.

Relation to Intelligent Agents

Simple Reflex

Reacts only to immediate surroundings. Cannot remember explored paths. Performs poorly in unknown mazes.

Model-Based

Stores visited locations. Builds an Internal map.

Goal-Based

Goal: Reach the exit. Chooses actions that moves forward to the goal.

Utility-Based

Considers path cost, distance, battery usage and safety.

Effects of Agent Design and decision-Making.

The design of the Agent directly influences its decisions.

Best Combination

The most effective combination is a Model-Based Utility Agent with UCS.

Justification

The Model-Based Agent remembers explored paths and updates its internal map.

The Utility-Based chooses actions that maximize safety, minimize energy usage, and reduce travel cost.

Uniform Cost Search guarantees the least-cost path when movement costs vary.

For an unknown maze, uniform Cost Search is the best choice when finding the minimum cost path is important, whereas IDS is preferred in memory-constrained situations.